Quantum-Secured Optical Backhaul Networks for 6G Communications: A BB84-Based FSO–Fiber Performance Study

Vriti Miglani, Arav Jain, Yugnanda Malhotra, Jolly Parikh
Bharati Vidyapeeth's College of Engineering, New Delhi

"Beyond classical communication: securing 6G with quantum light."

ABSTRACT

A quantum-photonic convergence framework integrating BB84-QKD, hybrid FSO–fiber propagation, and attenuation aware performance analytics is proposed for secure 6G optical infrastructures. The architecture enables AI-assisted adaptive routing, predictive channel resilience, and quantum-aware intrusion detection across next-generation post-classical communication fabrics.

WHY IT MATTERS?

- **6G optical backhaul needs both very high throughput and provably secure key exchange between nodes.**
- **QKD (BB84) gives information-theoretic key security via polarization-encoded single photons — but photons are fragile.**
- **FSO offers fast, license-free wireless optical links, but the open channel is exposed to haze, rain, fog and turbulence.**

OUR APPROACH: System Architecture

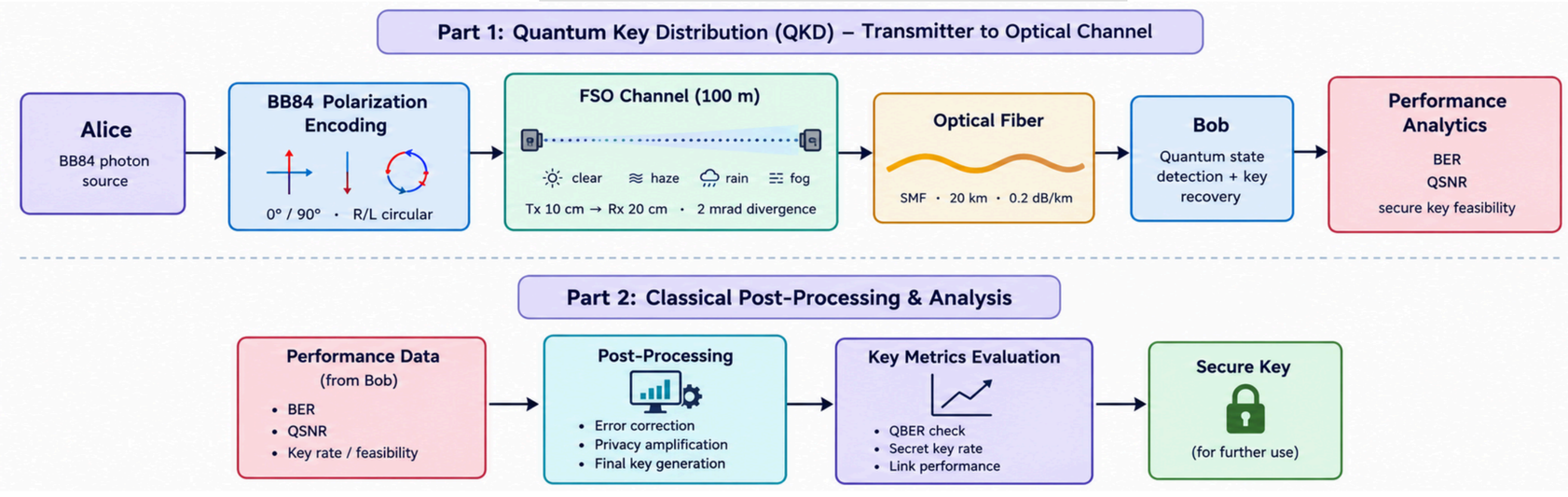
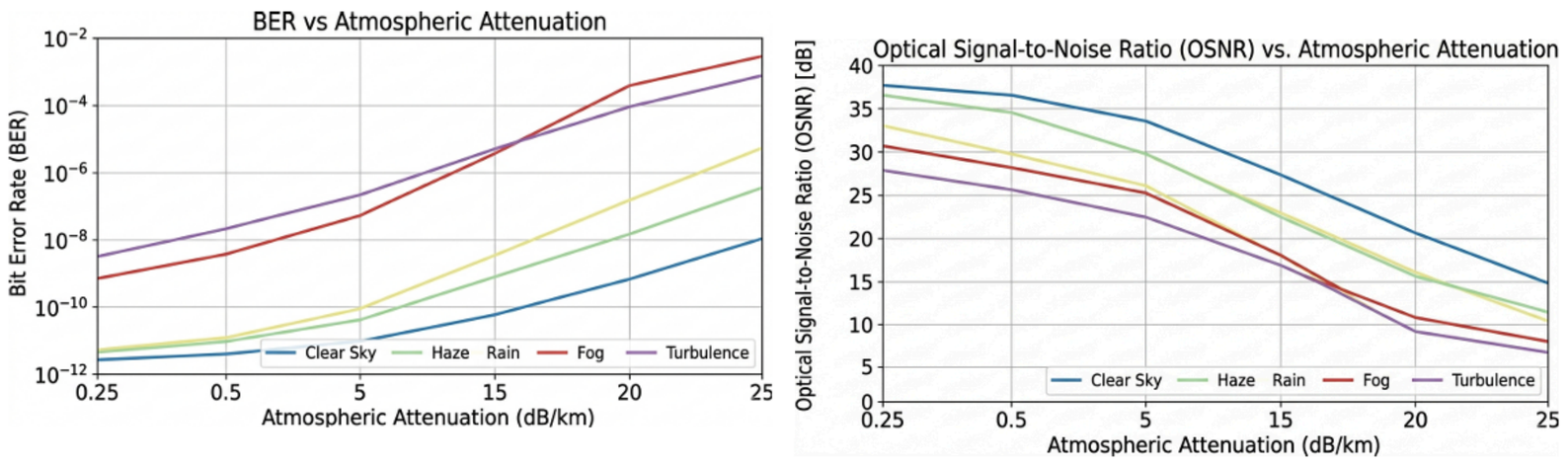


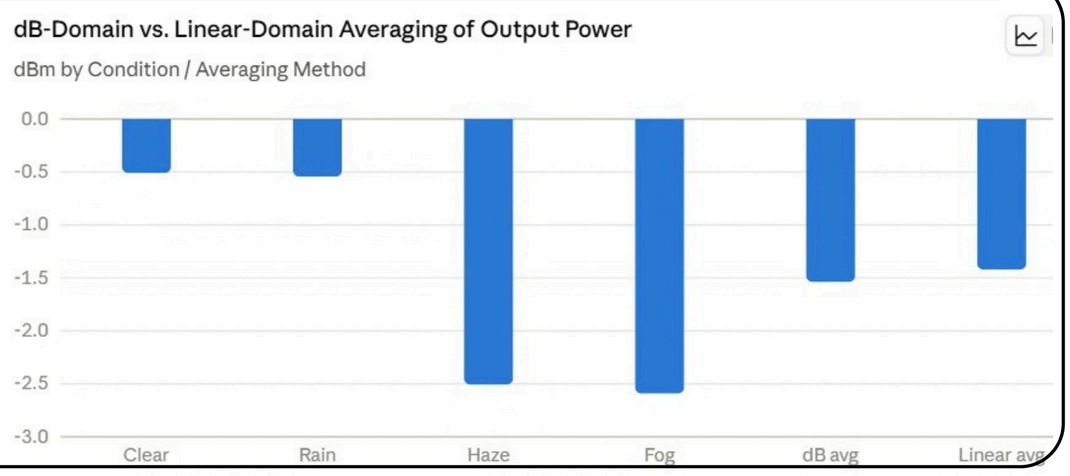
Fig 1: Atmospheric impairments: turbulence ,scintillation , haze , rain ,fog attenuate the FSO hop before the signal re-enters guided fibre

GRAPHICAL ANALYSIS



- **BER rises by ~6–8 orders of magnitude as attenuation increases from 0.25 to 25 dB/km, while OSNR falls by over 25 dB across the same range — confirming attenuation as the single dominant driver of channel degradation.**
- **Clear sky consistently outperforms all other conditions on both metrics, while fog and rain converge to the worst-case BER/OSNR at high attenuation — setting the practical operating limit for secure key exchange.**

Is the reported average link performance computed in the dB domain or the linear (transmission) domain — and does that choice shift the conclusion?"



METHODOLOGY PIPELINE

BB84 photon generation→Polarization encoding→FSO transmission→Atmospheric attenuation→Fiber transmission→Optical detection→ OSNR + BER analysis→Channel performance /key feasibility

CONCLUSION

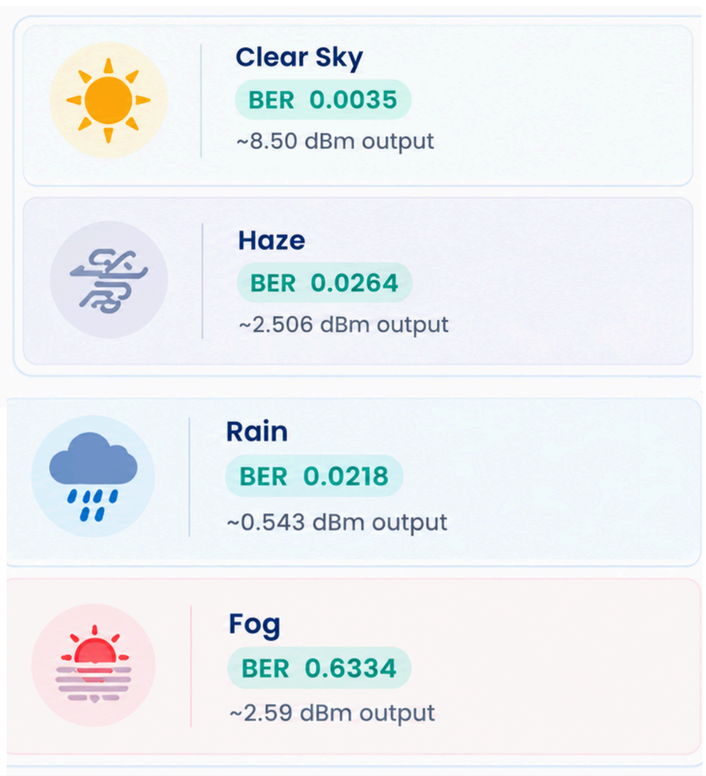
Averaging performance in the dB domain slightly understates true link quality : for this dataset, the reported average output power drops by ~0.12 dB compared to the physically correct linear-domain average. The gap is modest here because the four conditions aren't extreme, but the direction confirms dB averaging systematically biases results toward the worst-case condition, reinforcing that link-budget reporting should default to linear-domain (transmission-weighted) averaging for physical accuracy.

FUTURE SCOPE

Going forward, there's a plan to extend this averaging check across a continuous range of atmospheric attenuation values rather than just four discrete conditions, to better quantify how the gap between dB-domain and linear-domain averaging grows under more extreme or rapidly varying weather. There is also an idea to intend to apply the same transmission-domain averaging principle to secure key rate reporting, not just output power, to make sure my QKD performance claims aren't similarly understated by log-scale bias.

WEATHER SRESS TEST

- **Clear skies keep the link crisp — fog turns it chaos. Same laser, wildly different fate.**
- **From haze to fog, BER climbs nearly 10× — the atmosphere is the real adversary here, not the eavesdropper.**
- **Every raindrop costs you signal. Every dB of loss costs you key.**



RESULTS

Performance under different Atmospheric Condition

Condition	BER	Output Power (dBm)
Clear Sky	0.0035	-0.509
Rain	0.0218	-0.543
Haze	0.0264	-2.506
Fog	0.0334	-2.590

- **BER rises with attenuation, cutting directly into secure-key generation efficiency.**
- **The BB84-enabled hybrid FSO–fiber framework holds up across all four weather conditions, supporting its use for 6G optical backhaul study.**

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